

Insufficient Evidence Report: Lack of Biological Data for Identifying Human-Specific Genes

Input Question

What genes make us uniquely human?

Domain

Biology

Validation Status

needs_data

Problem Statement

The question seeks to identify specific genetic sequences or differences that distinguish the human genome from that of other primates (specifically chimpanzees) and determine which of these differences are associated with the evolution of the human brain. However, the provided evidence catalog contains no biological data.

Rationale

The allowed evidence IDs (EV-Q029-5f275837128937a833c7e4c5, EV-Q029-67f2b11ff24f0c4c84d0e37a) pertain exclusively to Human-AI Interaction and gesture recognition technologies. They contain zero information regarding genomics, evolutionary biology, or neuroscience. Consequently, it is impossible to derive factual claims or validated hypotheses about human-specific genes from the provided evidence. This report explicitly declares a knowledge gap and refuses to fabricate biological facts.

Generated Hypotheses

Technical Details

由于允许的 evidence_ids (EV-Q029-5f275837128937a833c7e4c5, EV-Q029-67f2b11ff24f0c4c84d0e37a) 仅涉及人机交互与手势识别，完全缺乏关于人类基因组、灵长类比较或神经进化的生物学数据，因此无法基于现有证据构建针对'人类独特基因'的科学假设。本计划旨在设计一个独立的计算基因组学研究框架以填补知识空白，但明确声明当前阶段仅为方法论提案，不包含基于给定证据的事实断言，且所有生物学相关主张均标记为 knowledge_gap。

Datasets

Source

```json

```
[
{
"name": "Human Genome Reference (GRCh38)",
"description": "Baseline human genomic sequence for identifying conserved and divergent regions.",
"access_note": "Publicly available via NCBI/Ensembl; not yet downloaded in this session."
},
{
"name": "Chimpanzee Genome Reference (Pan_tro_3.0 or later)",
"description": "Comparative primate genome to identify human-specific substitutions and structural variants.",
"access_note": "Publicly available via NCBI/Ensembl; not yet downloaded in this session."
}
]
'''
```

## Target

```
'''json
{
"name": "Human Accelerated Regions (HARs) and Human-Specific Gene Duplications",
"description": "Candidate genomic elements showing significant divergence in humans compared to chimpanzees and other primates, potentially linked to brain development."
}
'''
```

## Paper Abstract

Background: The question asks for genes that make humans unique compared to primates. Methods: We reviewed the provided evidence catalog (EV-Q029-5f275837128937a833c7e4c5, EV-Q029-67f2b11ff24f0c4c84d0e37a). Validation Plan: Attempted to extract genetic data. Results: Pending/Failed. The provided evidence is unrelated to biology (covering HCI and gesture recognition). Therefore, no hypotheses could be generated, and no results were obtained. Conclusion: The question cannot be answered with the current evidence set; it represents a critical knowledge gap.

## Methods

Evidence Review and Gap Analysis. No biological experiments were conducted. No computational genomics analyses were performed. The methodology is limited to identifying the mismatch between the research question

(Biology) and the available evidence (Computer Science/HCI).

## Experiments

### Baselines

```
```json
[
  "Neutral Evolution Model: Assumes constant mutation rates across lineages without selection pressure.",
  "Conserved Non-coding Elements (CNEs) from Mouse-Human Comparison: Standard baseline for functional regulatory elements."
]
```
```

### Metrics

```
```json
[
  "Substitution Rate Ratio (Human/Chimp vs. Outgroup)",
  "PhyloP Score for Conservation/Acceleration",
  "Overlap Significance with Brain-specific Enhancers (p-value)"
]
```
```

### Ablation

Exclude regions overlapping with known repetitive elements to reduce false positives in alignment.

### Validation Protocol

Cross-validate identified HARs against independent datasets of fetal brain chromatin accessibility (e.g., ATAC-seq data from public repositories, not provided in current evidence).

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- **\*\*EV-Q029-5f275837128937a833c7e4c5\*\*** · arxiv · arXiv:2603.12895
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2603.12895.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:7|section:page-7|paragraph:1;  
content\_sha256=f963ceb9fd93cdc0159f76b94b76b65f407e93b81c92715f7722be5024237a46
- **\*\*EV-Q029-67f2b11ff24f0c4c84d0e37a\*\*** · arxiv · arXiv:1909.06415
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1909.06415.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:4|section:page-4|paragraph:1;  
content\_sha256=f12435428ed5921413e14c1f16b11bc5885af601b06b8a1e5e8296f588988309

## Reviewer Comments

- The submission correctly maintains the 'insufficient\_evidence' status and refuses to fabricate biological hypotheses based on irrelevant HCI evidence cards (EV-Q029-5f275837128937a833c7e4c5, EV-Q029-67f2b11ff24f0c4c84d0e37a).
- No results were fabricated; execution metadata explicitly states actual\_execution=false due to domain mismatch.
- Knowledge gaps are formally declared for human-specific genetics and brain evolution, adhering to previous revision requirements.
- However, the plan still fails the mandatory pass condition 'at least 1 hypothesis must be falsifiable' because no valid hypothesis could be generated from the allowed evidence set.
- The proposed comparative genomics framework remains a methodological suggestion for future data acquisition rather than a testable hypothesis grounded in current evidence.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Define exact genome assembly versions for Human and Chimpanzee.
- Specify alignment tool parameters (e.g., LASTZ, MULTIZ).
- List statistical thresholds for identifying acceleration (e.g., FDR < 0.05).
- Document code for phylogenetic modeling.